

Asen Dotsinski

+359888870770
asendotsinski@protonmail.com
Amsterdam, The Netherlands
Bulgarian



SUMMARY

I am a quick learner, ambitious to a fault and I never back down from a challenge. I am looking to use my AI research talent for the good of humanity. I am currently interested in the mechanisms behind LLM prompt attacks and counteracting them to safeguard (open-weight) AI against malicious use.

EDUCATION

- | | |
|---|-------------------------------|
| University of Amsterdam | Amsterdam, Netherlands |
| <ul style="list-style-type: none">♦ Master of Science – MS, Artificial Intelligence♦ Average grade: 8.9/10♦ Main author of a reproducibility study in LLM interpretability¹ and co-author of a paper on deep-fake detection²♦ Finalizing a paper in which a jailbreaking method achieves up to 97% ASR with one line of code | 2024 – 2026 |
| Eindhoven University of Technology | Eindhoven, Netherlands |
| <ul style="list-style-type: none">♦ Bachelor of Engineering – BE, Computer Science♦ Average grade: 9.2/10♦ Key courses: Honors Research Project, Process Mining, Responsible Machine Learning | 2021 – 2024 |
| High School of Mathematics “Doctor Petar Beron” | Varna, Bulgaria |
| <ul style="list-style-type: none">♦ Key modules: IT, Software Development, Mathematics, English | 2016 – 2021 |

EXPERIENCE

- | | |
|--|-----------------------|
| Freelance machine learning engineer [C++, Python, PyTorch, CUDA, Docker] | Remote |
| Motive Engineering | 2024 – present |
| <ul style="list-style-type: none">♦ Implemented a swarm optimization algorithm entirely on CUDA, achieving a 17x speedup over previous solutions while halving the lines of code used♦ Created a combined DevOps/MLOps CD flow using Docker and pre-commits, reducing deliverable size tenfold and ensuring code reproducibility♦ Designed, implemented and optimized multiple object collision algorithms♦ Refactored legacy code to improve maintainability and use C++ 17 best practices | |
| Back-end developer [Ruby on Rails, Rspec, PostgreSQL, TypeScript] | Eindhoven |
| Drieam | 2023 – 2024 |
| <ul style="list-style-type: none">♦ Maintained and expanded the backend of a Rails Ed-Tech application integrated with Canvas♦ Applied test-driven development using Rspec and wrote API contracts in OpenAPI | |
| Embedded developer [C/C++, Bash, FreeRTOS, Jira, Scrum] | Best |
| WeFabricate | 2022 – 2023 |
| <ul style="list-style-type: none">♦ Designed and developed real-time firmware for a fleet of 150 embedded IoT controllers of dis- | |

¹ Dotsinski, Asen, et al. "On the Generalizability of 'Competition of Mechanisms: Tracing How Language Models Handle Facts and Counterfactuals'." TMLR 2835-8856 (2025) ([arXiv:2506.22977](https://arxiv.org/abs/2506.22977))

² Thakur, Udit, et al. "CLaRE: CLIP with Latent Reconstruction Errors for Generated Face Detection." Proceedings of the 1st ACM Workshop on Deepfake, Deception, and Disinformation Security. 2025.

persing systems for international clients like Walmart and Lidl

- ◆ Implemented live data analysis and calibration of sensors, with automatic error correction
- ◆ Configured and maintained a production-ready Linux-based environment using Bash scripts
- ◆ Led and managed a small software team (3-5 people) using Jira and Scrum

C++ developer [C++, Git]

Veldhoven

Dimenco

2021 – 2022

- ◆ Developed and tested a C++ library for automating calibration tasks with an industrial robotic arm

Robotics teacher, content developer [LEGO Mindstorms]

Bulgaria

Robopartans

2019 – 2021

- ◆ Taught children aged 7-15 how to build and program robots using the LEGO EV3 Mindstorms kit
- ◆ Developed materials and lessons for the online curriculum the company uses at filcasts.com

EXTRACURRICULAR

Founder, president and trainer

Varna, Bulgaria

Debate club “Odessos”

2018 – 2021

- ◆ Founded the only active debate club in the region, trained and coached multiple national and international winners

Founder, lead engineer

Varna, Bulgaria

Laboratory for Innovative Technology (LIT)

2017 – 2019

- ◆ Founded a student team that introduces students to Arduino programming and hardware technologies, while also creating solutions for local businesses and governments

Competitive C++ programmer

Varna, Bulgaria

High School of Mathematics “Doctor Petar Beron”

2013 – 2019

- ◆ Participated in over 30 national and international C++ coding competitions
- ◆ Received 2 sponsored invitations for the National Informatics Training Camp
- ◆ Finished twice in the top 10 national yearly rankings

SPONSORSHIPS & AWARDS

- ◆ 7 certificates of excellence and multiple top-3 placements in national informatics competitions
- ◆ 1st place in the international science fair “Ot vinta!” in 2018
- ◆ “Judges’ Choice Award” in Junior Achievements’ competition “Rising Stars” in 2019
- ◆ Over 10 top-3 placements in national debate competitions

LANGUAGE SKILLS

- ◆ Bulgarian (native tongue)
- ◆ English (C2) – IELTS Overall Band Score 8.5/9
- ◆ Russian (B1)

HOBBIES

- ◆ Reading up on AI research, especially regarding AI capability and safety
- ◆ Playing board games, especially Netrunner
- ◆ Spending time with my extended family and friends